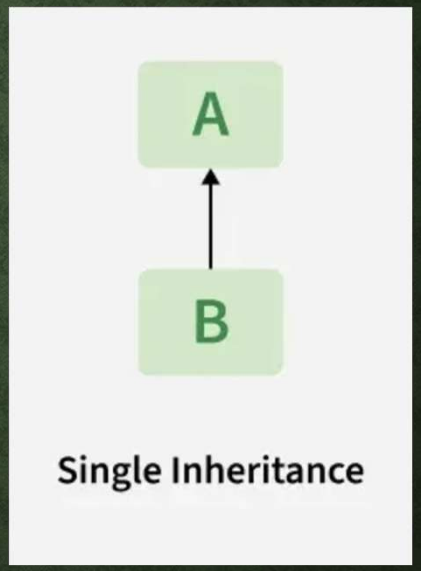
INHERITANCE

- One of the core concepts of Object-Oriented Programming (OOP)
- allows a derived class/child class) to inherit properties (fields) and behaviors (methods) from a base class/parent class.
- promotes code reusability and extensibility

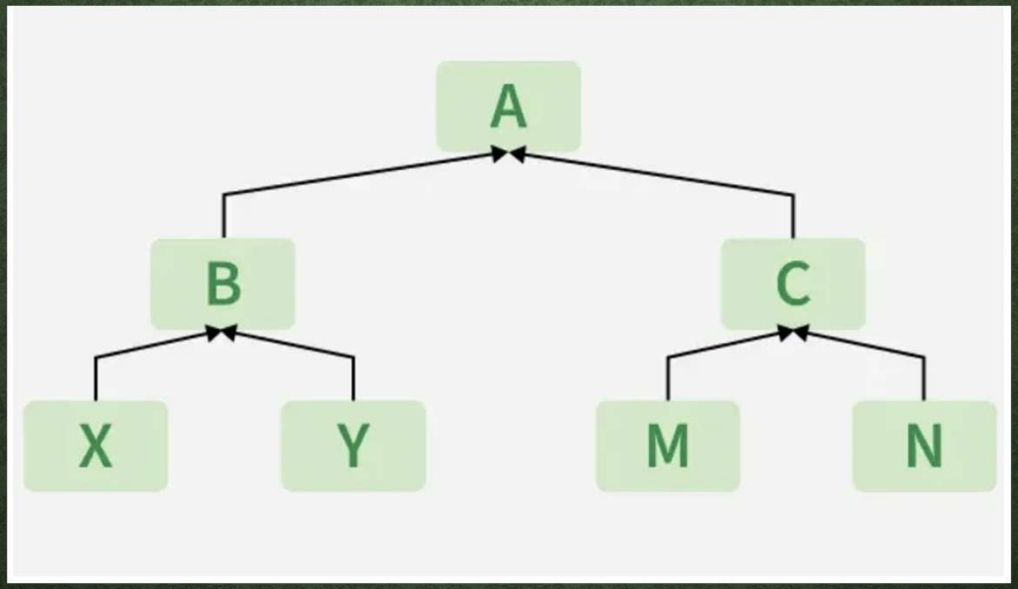
SINGLE INHERTIANCE

```
class BaseClass
{
        // Members of base class
}

class DerivedClass : BaseClass
{
        // Members of derived class
}
```



HIERARCHICAL INHERITANCE



CONSTRUCTOR

- a derived class's constructor can explicitly call a base class constructor using the base() keyword.

- Base class's functionalities reused by multiple derived classes, reducing code duplication.
- Derived classes can extend the functionality of the base class by adding new members or overriding existing virtual methods

```
class Animal {
    public void Eat() {
        Console.WriteLine("This animal eats food.");
    }
}

class Dog : Animal {
    public void Bark() {
        Console.WriteLine("The dog barks.");
    }
}
```

```
class Program {
    static void Main() {
        Dog d = new Dog();
        d.Eat();  // Inherited method
        d.Bark(); // Derived class method
    }
}
```

VIRTUAL

virtual keyword modifies a method, property, indexer, or event declaration in a base class, allowing it to be overridden in a derived class.

POLYMORPHISM

- When you have a base class reference pointing to a derived class instance, calling the virtual member will execute the derived class's overridden implementation.
- This is known as runtime polymorphism.

```
class Animal {
    public virtual void Eat() {
        Console.WriteLine("This animal eats food.");
    }
}
class Dog : Animal {
        public override void Eat() {
        Console.WriteLine("This dog eats food.");
    }
        public void Bark() {
        Console.WriteLine("The dog barks.");
    }
}
```

```
class Program {
    static void Main() {
        Dog d = new Dog();
        d.Eat();  // Inherited method
        d.Bark(); // Derived class method

        Animal a = new Dog();
        a.Eat();  // This dog eats food
        //a.Bark(); // error

    }
}
```